

AI-Powered Lead Generation System

Salesforce Certified Administrator with AI Agentforce Specialization

Submitted By:

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Project Overview

The **AI-Powered Lead Generation System** was developed as part of the **Salesforce Certified Administrator with AI Agentforce Specialization Virtual Internship** using **Salesforce Developer Edition**. The primary objective of this project was to design and configure a secure, intelligent, and efficient Lead Management solution that streamlines the process of capturing, organizing, and managing sales leads within the Salesforce platform. The project demonstrates the practical application of Salesforce Administration concepts while integrating AI capabilities to enhance productivity and decision-making.

The solution includes customization of the standard Lead object through the creation of custom fields, page layouts, validation rules, and automation using Salesforce Flow. Role-based security was implemented by configuring custom **Profiles, Users, Roles, and Permission Sets**, ensuring that users have appropriate access based on their responsibilities. These configurations help maintain data integrity, improve security, and provide controlled access to organizational information.

A key feature of this project is the integration of **Salesforce Agentforce Prompt Builder**, which enables the generation

of AI-powered Lead summaries directly within Lead records. By combining Salesforce Administration with generative AI, the system reduces manual effort, improves the quality of Lead information, and enhances the overall user experience. The completed solution demonstrates how modern Salesforce technologies can be used to build a scalable, secure, and business-oriented CRM system that supports efficient sales operations and informed decision-making.

Objectives

- The AI-Powered Lead Generation System was developed with the following objectives to improve Lead management, strengthen data security, and enhance business productivity within the Salesforce platform:
- **Enhance Lead Management:** Customize the standard Salesforce Lead object to efficiently capture, organize, and manage customer information throughout the sales process.
- **Improve Data Quality:** Implement validation rules to ensure that only accurate and relevant business information is entered into the system, reducing errors and maintaining data integrity.
- **Implement Role-Based Security:** Configure custom Profiles, Roles, Users, and Permission Sets to provide secure, role-based access while protecting sensitive business data.
- **Customize the Salesforce Environment:** Modify page layouts, custom fields, and object settings to create a user-friendly interface tailored to business requirements.
- **Automate Business Processes:** Design and implement Salesforce Flow automation to reduce manual effort, improve operational efficiency, and streamline Lead-related activities.
- **Integrate AI-Powered Lead Summaries:** Utilize Salesforce Agentforce Prompt Builder to generate intelligent Lead summaries, enabling users to quickly understand customer information and make informed decisions.
- **Strengthen CRM Security and Governance:** Apply Salesforce best practices for access control, data management, and system configuration to ensure a secure and scalable CRM solution.
- **Gain Practical Salesforce Administration Experience:** Demonstrate hands-on knowledge of Salesforce Administration by implementing real-world CRM configurations, automation, security, and AI features within a Salesforce Developer Edition environment.

Project Implementation

The implementation of the AI-Powered Lead Generation System was carried out using Salesforce Developer Edition by following Salesforce Administration best practices. The project began with configuring the Salesforce environment and enabling the required features. After the initial setup, the standard Lead object was customized by creating additional fields, modifying page layouts, configuring validation rules, and implementing automation. Security was strengthened by creating custom profiles, users, roles, and permission sets. Finally, Salesforce Prompt Builder was configured to generate AI-powered Lead summaries, and the complete solution was tested to ensure proper functionality and secure access.

Salesforce Environment Setup

The project started with logging into Salesforce Developer Edition and accessing the Lightning Experience interface. The Home page provides access to dashboards, recent items, and administrative tools, while the App Launcher allows navigation between Salesforce applications and features. This served as the starting point for configuring the complete Lead Management System.

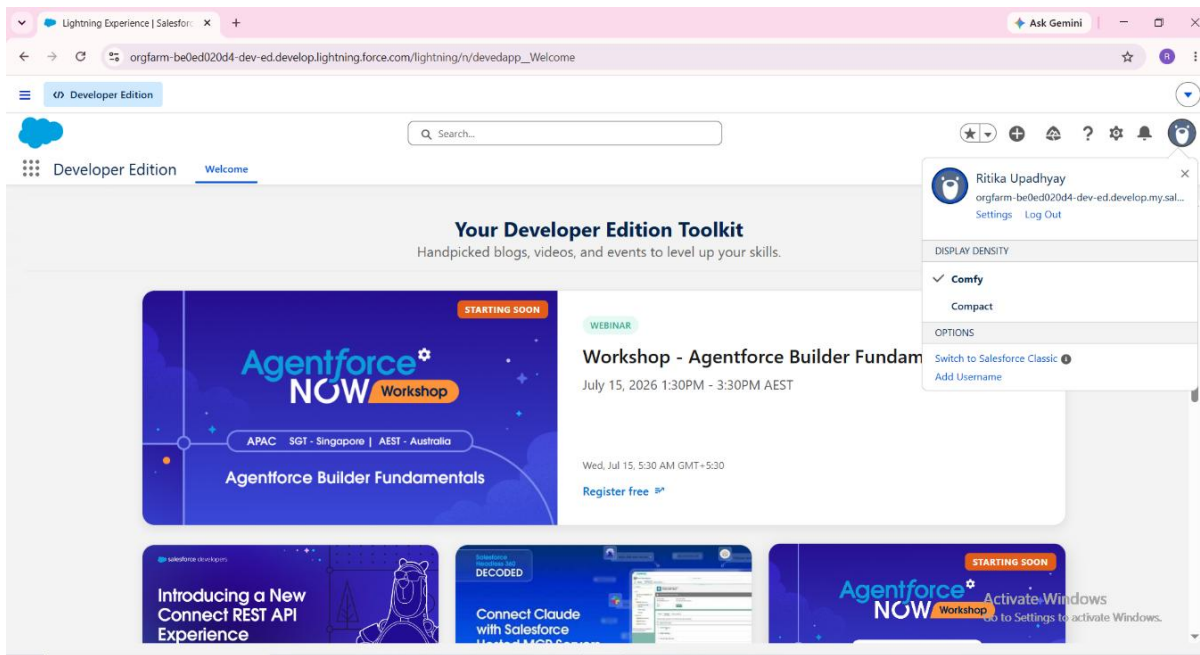


Figure 1: Salesforce developer Edition Home Page

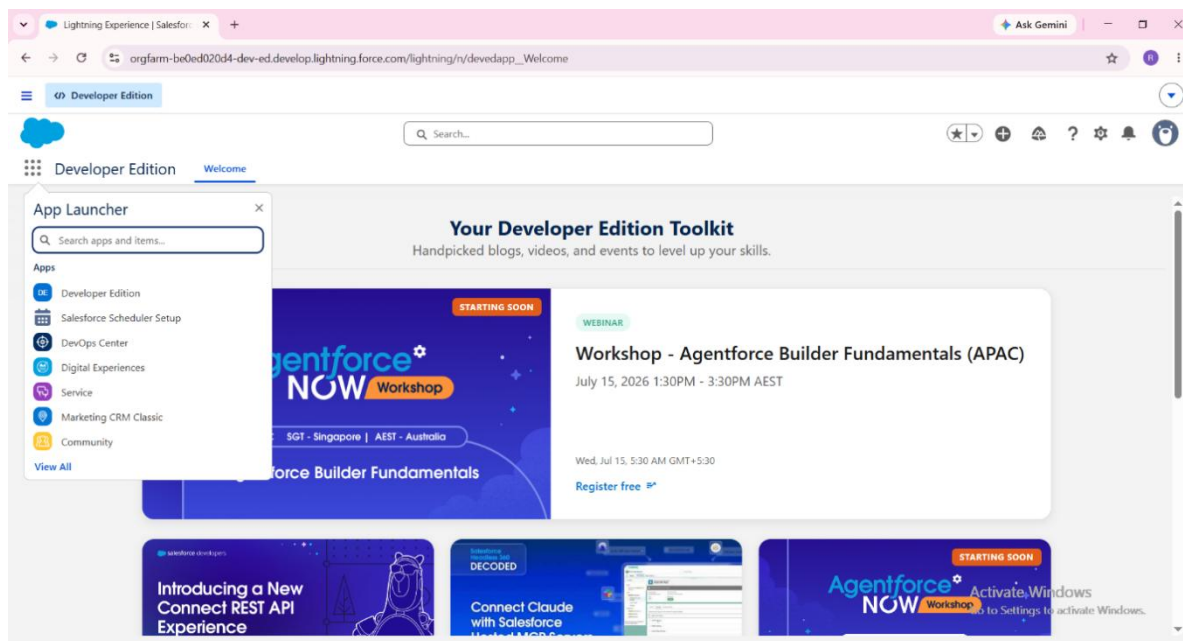


Figure 2: Salesforce App Launcher

Security configuration

To implement secure access control, custom profiles were created for different user responsibilities. The Lead Manager Profile provides broader administrative permissions, whereas the Sales Agent Profile is designed with restricted permissions to prevent unauthorized modification or deletion of records.

Profiles & Roles

To establish a secure and well-structured Salesforce environment, two custom profiles and a role hierarchy were configured based on the organization's requirements. These configurations ensure appropriate user access, protect critical business data, and maintain a clear reporting structure.

- **Lead Manager Profile & Role:** Granted full Create, Read, Edit, and Delete (CRUD) permissions on Lead records along with access to related Tasks and Contacts. The Lead Manager role provides visibility into records owned by subordinate users through the configured role hierarchy.
- **Sales Agent Profile & Role:** Granted Create, Read, and Edit (CRU) permissions while the Delete permission was restricted to prevent accidental removal of Lead records. This profile enables Sales Agents to manage their assigned Leads while maintaining data security and integrity.

The configured profiles and roles demonstrate Salesforce's role-based access control (RBAC) model, ensuring that users can access only the information and functionality required for their responsibilities.

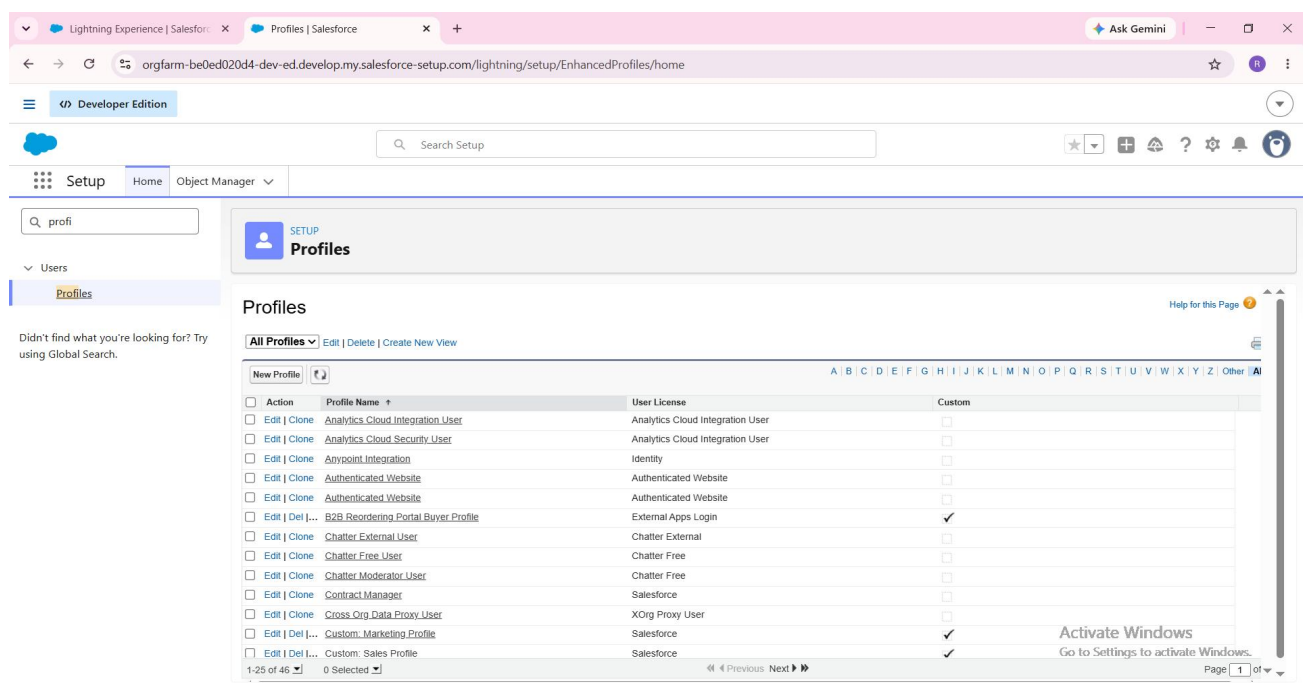


Figure 3: Profile List

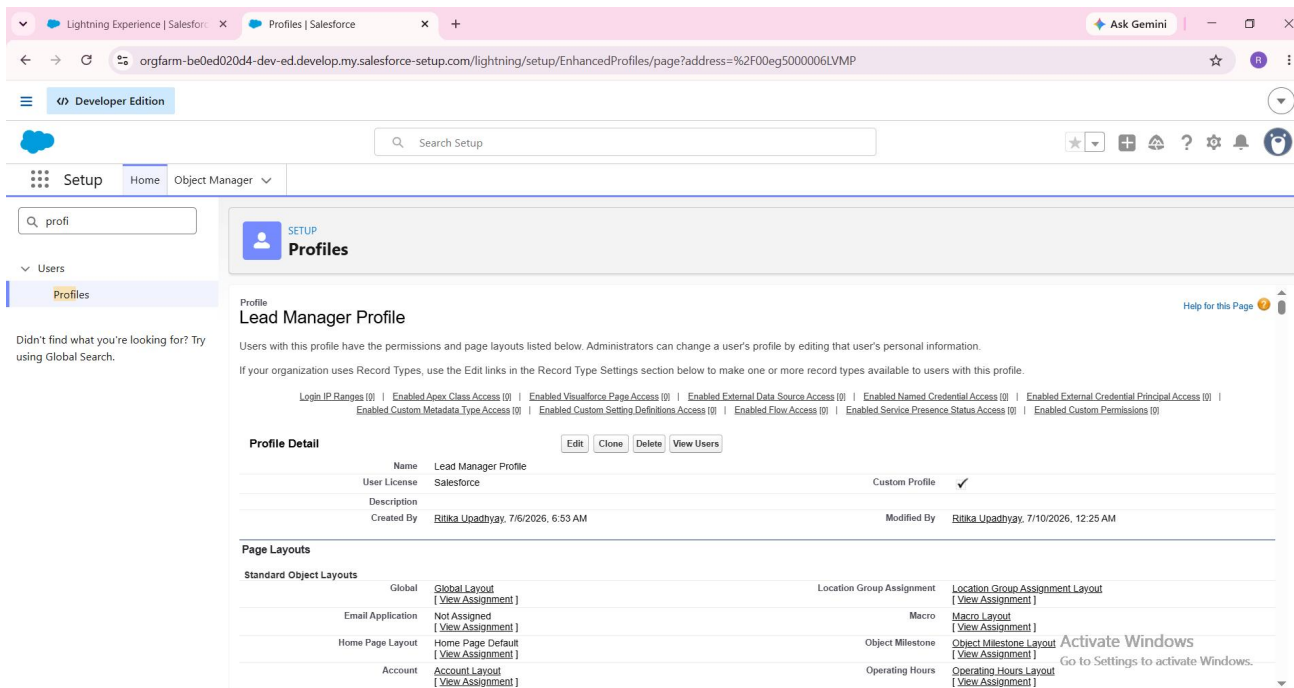


Figure 4: Lead Manager profile

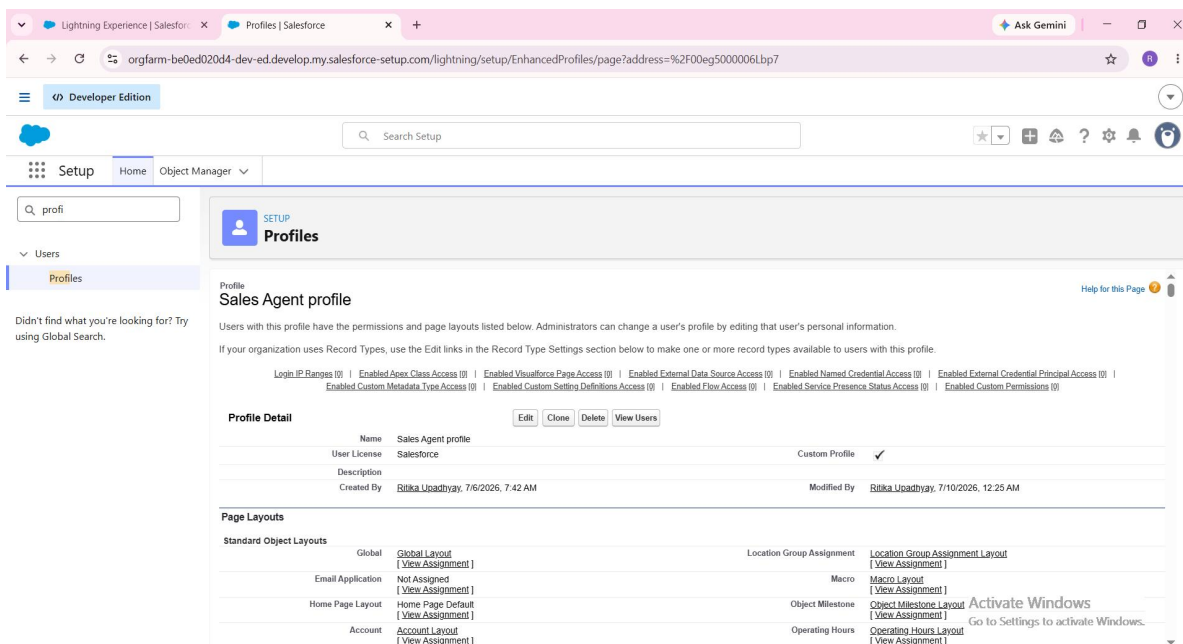


Figure 5: Sales Agent Profile

User Management

Two users were created to simulate a real organizational environment. Each user was assigned the appropriate profile and role to validate role-based security and ensure correct access to Lead records.

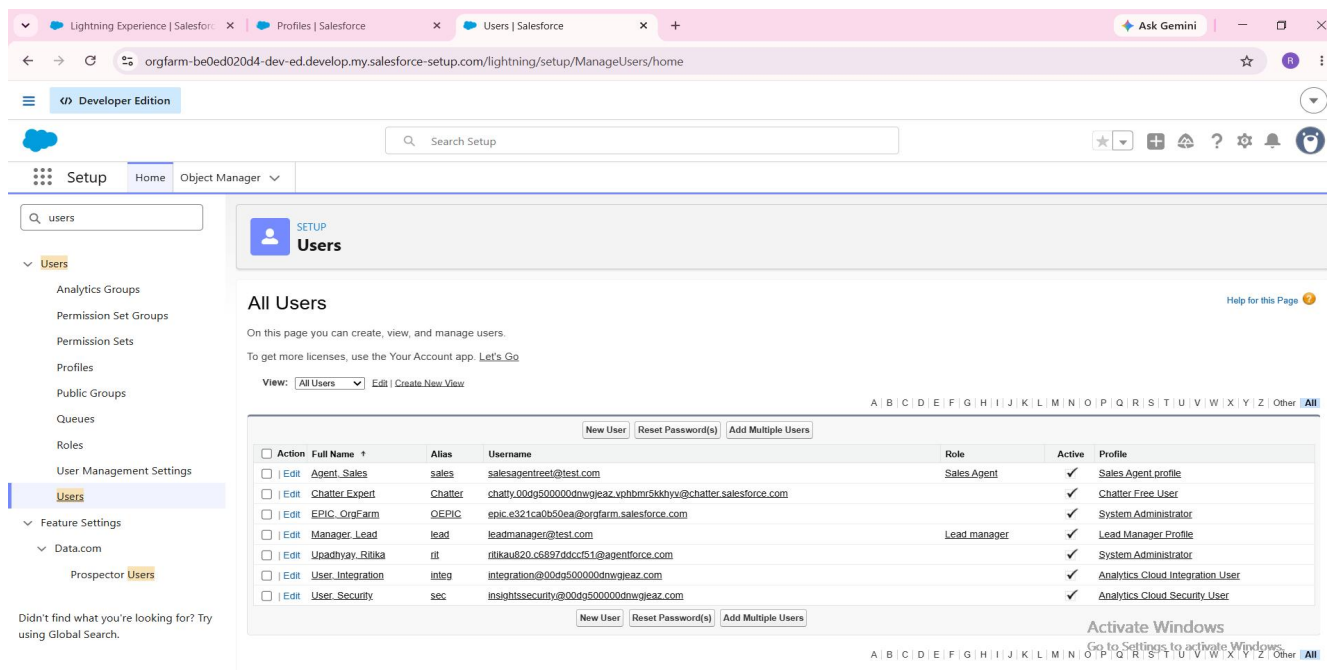


Figure 6: User List

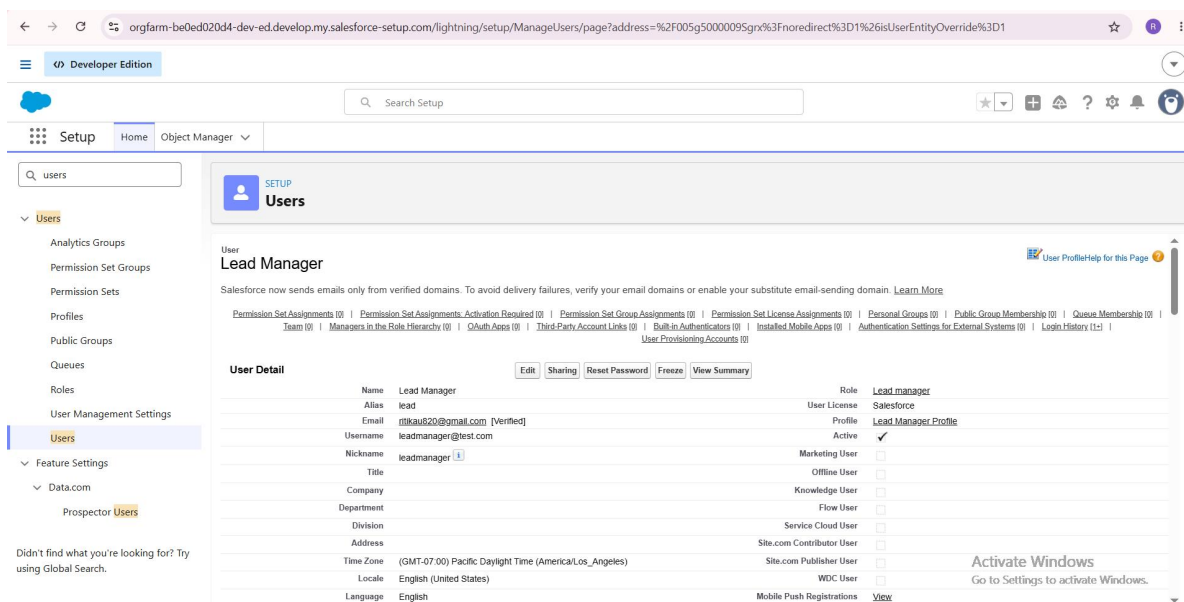


Figure 7: Lead Manager User Account

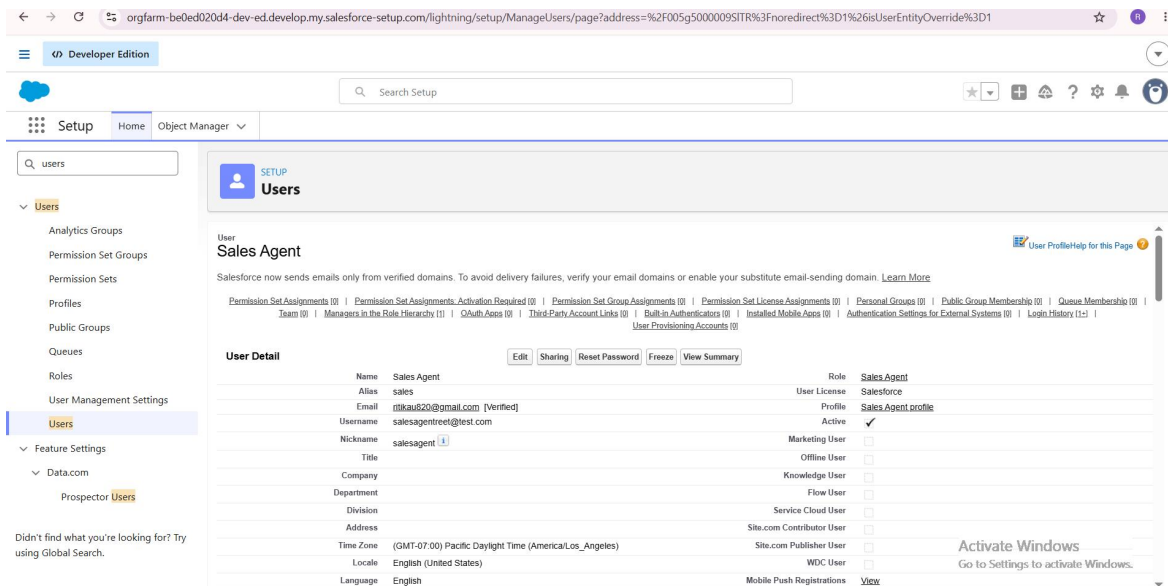
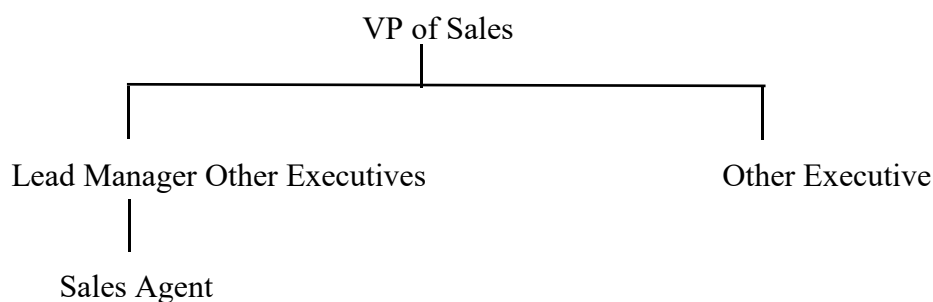


Figure 8: Sales Agent User Account

Role Hierarchy

A role hierarchy was configured to represent the organizational reporting structure. The Sales Agent reports to the Lead Manager, allowing managers to access records owned by their team while maintaining secure record visibility.



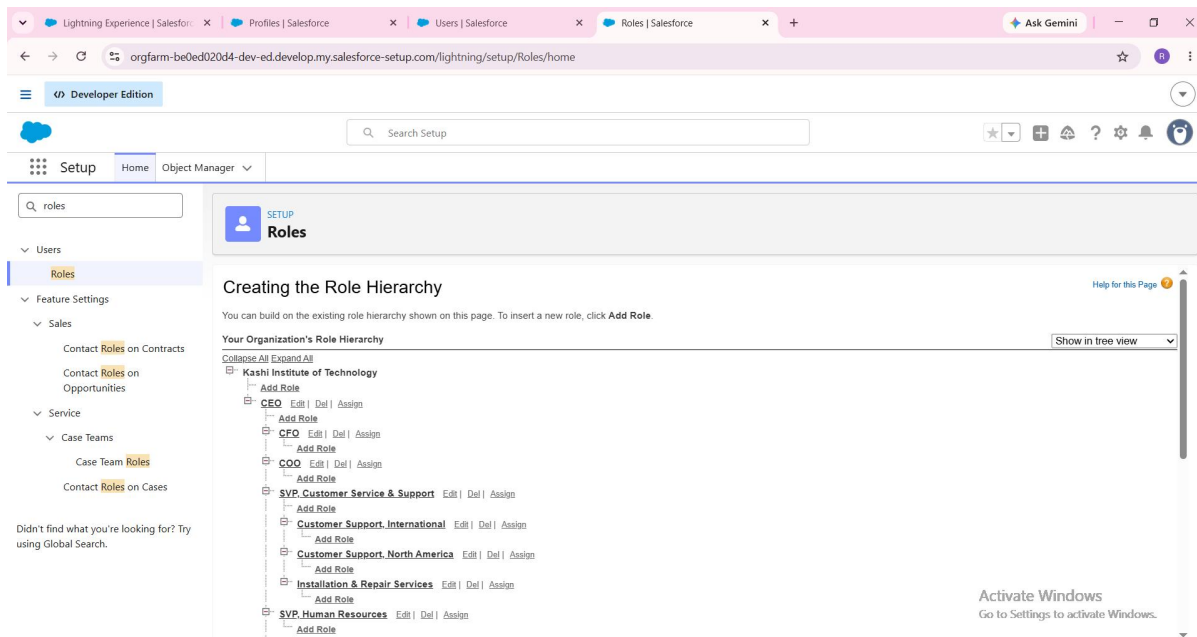


Figure 9: Role Hierarchy

Custom Fields

To support efficient Lead qualification and AI-powered automation, the standard Lead object was extended by creating custom fields based on the project requirements. These fields improve data organization, enable better Lead categorization, and provide a structured foundation for automated processing and AI-generated summaries.

The custom fields were created under **Object Manager → Lead → Fields & Relationships** with appropriate data types and configurations to capture essential business information.

The following custom fields were implemented:

Field Label	API Name	Data Type	Configuration Details
Lead Score	Lead_Score__c	Number	Length: 20, Decimal: 0
Lead Category	Lead_Category__c	Picklist	Values: Hot, Warm, Cold
Product Interest	Product_Interest__c	Picklist	Values: Training, Consulting, Certification
AI Summary	AI_Summary__c	Long Text Area	Length: 32,000, Visible Lines: 6

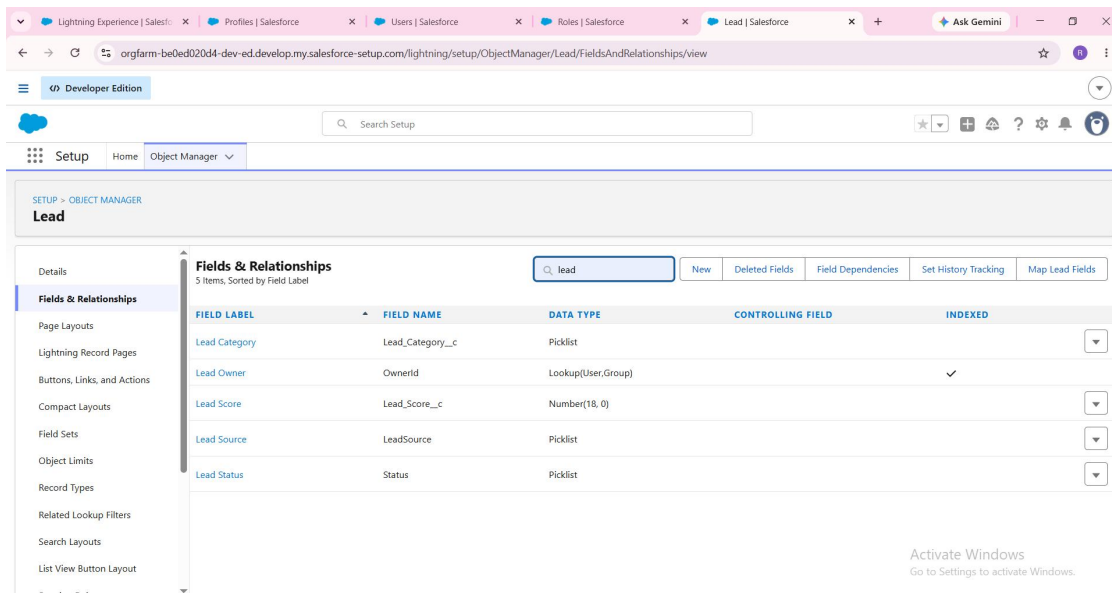


Figure 10: Lead Fields & Relationships

Lead Page Layout Customization

The Lead Page Layout was customized to organize important business information in a structured format. Frequently used fields were arranged logically to improve usability and provide sales representatives with quick access to relevant lead information.

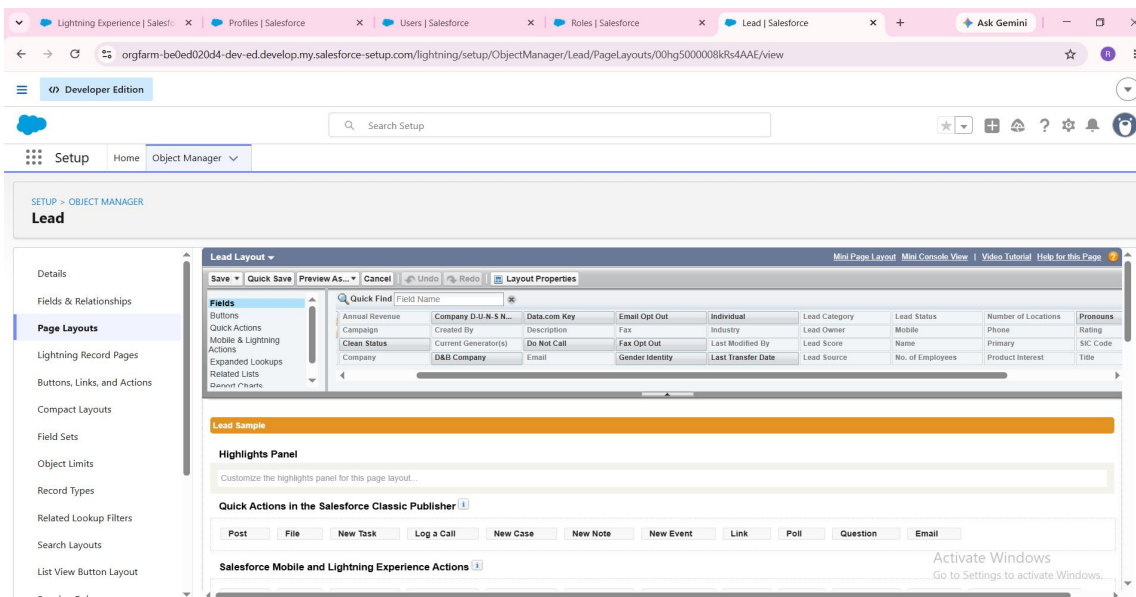


Figure 11: Lead Page Layout

Validation Rule

To maintain data quality and ensure that only valid business leads are captured, a custom Validation Rule named `Reject_Personal_Email` was implemented on the Lead object. The rule prevents users from saving Lead records that contain personal email domains such as Gmail, Yahoo, Hotmail, or Outlook.

When a restricted email domain is entered, Salesforce displays an appropriate error message and blocks the record from being saved until a valid business email address is provided. This validation helps improve the accuracy of Lead data and ensures that only qualified business contacts are stored within the system.

Error Message:

"Personal email domains like Gmail, Yahoo, Hotmail, and Outlook are not allowed." **Error**

Condition Formula Logic:

```
OR (
  CONTAINS (Email, "@gmail."),
  CONTAINS (Email, "@yahoo."),
  CONTAINS (Email, "@hotmail."),
  CONTAINS (Email, "@outlook.")
)
```

Flow Automation

To automate the lead management process, a Record-Triggered Flow was configured to execute automatically whenever a Lead record is created or updated. The flow is optimized using the Actions and Related Records option, enabling background automation without requiring manual intervention.

The automation includes a Send Email action that retrieves the Lead's email address dynamically and sends an automated notification. The email incorporates key Lead information along with the generated AI Summary, ensuring that users receive relevant and contextual information immediately after record processing. This workflow reduces manual effort, improves communication, and streamlines the overall lead management process.

Lead Record Testing

After completing the system configuration, sample Lead records were created to verify the functionality of all implemented features. Testing confirmed that custom fields, page layouts, validation rules, profiles, roles, flow automation, and AI-powered summary generation were functioning as expected.

The testing process demonstrated that the system successfully validates business email addresses, executes automated flows, generates AI summaries using Prompt Builder, and enforces the configured security permissions, ensuring a reliable and efficient Lead Management solution.

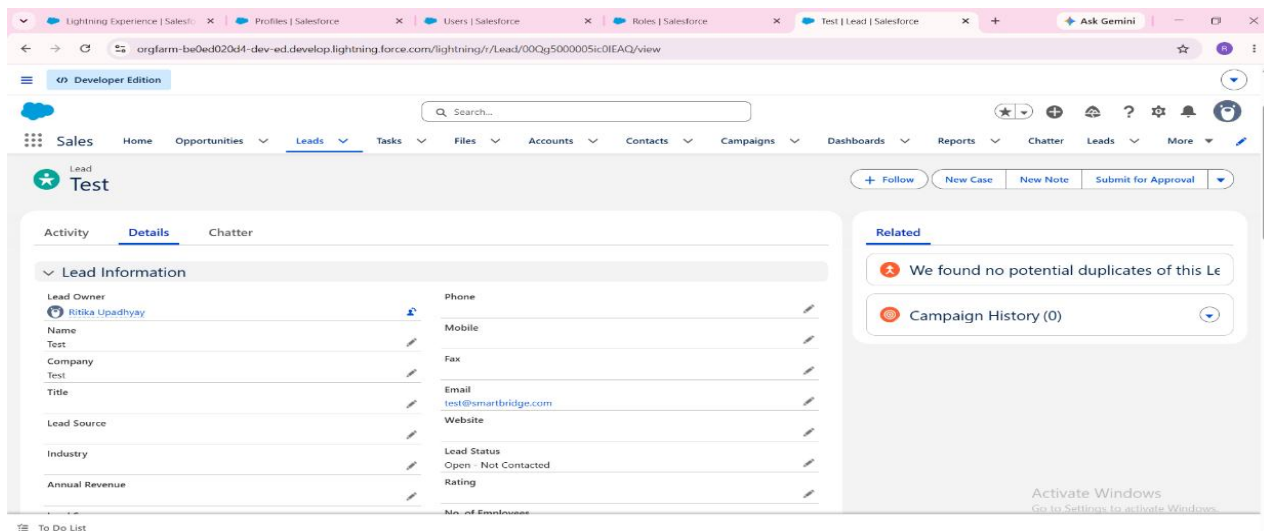


Figure 12: Lead record testing

Agentforce Prompt Builder

To enhance the Lead Management System with AI capabilities, Salesforce Agentforce Prompt Builder was configured to generate intelligent summaries for Lead records. A Field Generation Prompt Template was created and linked to the custom AI_Summary__c field to automate the generation of concise and structured Lead summaries.

The prompt template was designed to use dynamic Lead record information, including the Lead Name, Company, Email, Phone Number, Lead Source, Status, and Description. These fields provide the necessary context for generating meaningful and accurate summaries based on the available CRM data.

The following prompt instruction was configured within Prompt Builder:

"Generate a detailed summary of this Lead using the selected fields. Explain the Lead's background, company information, contact details, status, source, and any key notes clearly. Summarize relevant details such as current engagement stage and potential interest level in a clean, structured format."

After the prompt template was saved and activated, it was integrated with the Lead Record Page using Lightning App Builder. The AI Summary field was configured to display the Prompt Builder action, enabling users to generate and save AI-powered Lead summaries directly from the Lead record with a single click.

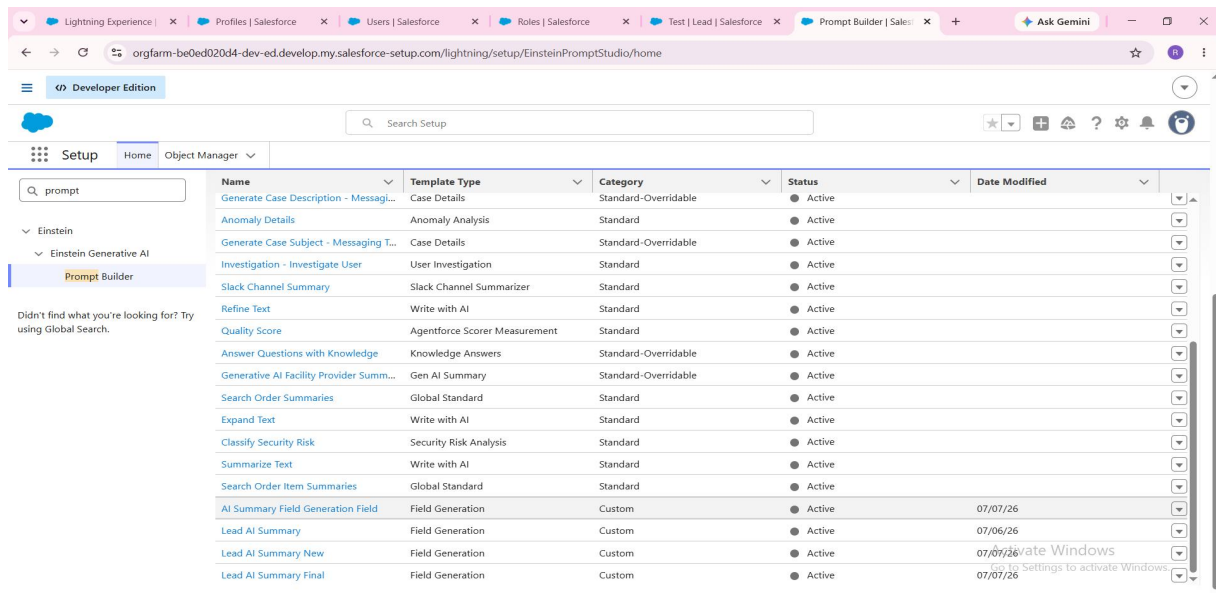


Figure 13: Agentforce Prompt Builder

Results

The implemented Lead Management System was successfully tested using sample Lead records within the Salesforce Developer Edition environment. All major configurations, including Lead object customization, custom fields, validation rules, profiles, users, role hierarchy, permission sets, Flow automation, and Agentforce Prompt Builder, functioned as expected.

Testing confirmed that the validation rules correctly restricted invalid data entry, the automated Flow executed successfully when the defined conditions were met, and the Prompt Builder generated AI-powered Lead summaries based on the configured Lead information. The generated summaries were successfully stored in the AI Summary field, demonstrating seamless integration between Salesforce Prompt Builder and the Lead Record Page.

Overall, the project met its intended objectives by providing a secure, well-structured, and AI-enabled Lead Management System that improves data quality, enhances user productivity, and streamlines Lead management processes.

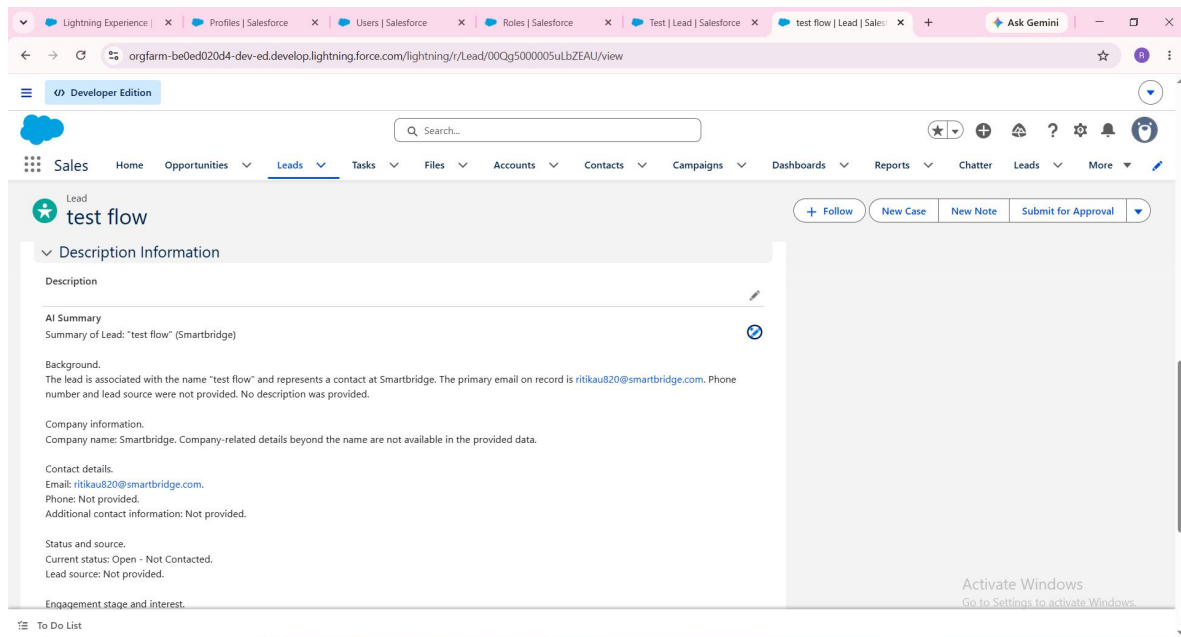


Figure 14: AI-Generated Lead Summary

Conclusion

- This project demonstrates the practical application of Salesforce Administration and Agentforce technologies to build a secure, organized, and AI-enabled Lead Management System. Throughout the project, core Salesforce features such as custom objects and fields, profiles, roles, permission sets, validation rules, Flow automation, and Lightning App Builder were successfully configured to create an efficient CRM environment.
- The integration of Salesforce Prompt Builder and AI-powered Lead Summary generation enhanced the user experience by enabling quick and meaningful insights directly within Lead records. Role-based security and access control ensured proper data governance while maintaining an effective organizational hierarchy.
- Completing this project provided valuable hands-on experience with Salesforce administration, security configuration, workflow automation, and AI integration. It strengthened my understanding of industry best practices and demonstrated how Salesforce can be customized to improve business processes, increase productivity, and support informed decision-making.
- The project has been successfully completed, tested, and documented as part of the Salesforce Certified Administrator with AI Agentforce Specialization Virtual Internship.

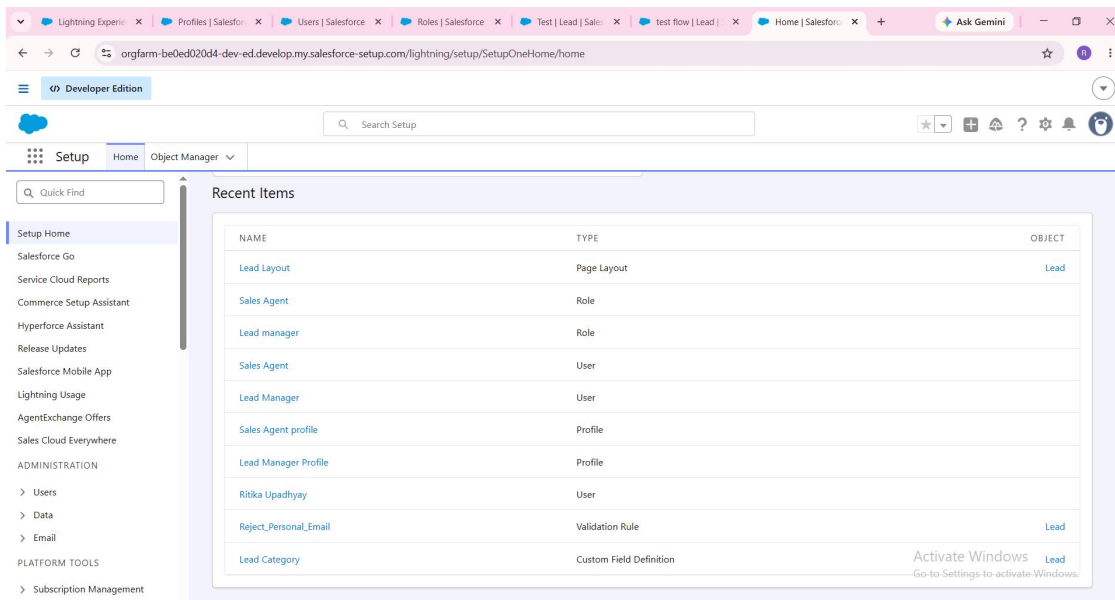


Figure 6: Final salesforce environment showing successful completion of the AI-Powered Lead Management project